Tracing Rationalistic/Scientific Outlook of Social Thinkers in 19th Century Bengal: A Narrative (Revised Version after Presentation)

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Scientific Method

"... science as a method of finding things out. This method is based on the principle that observation is the judge of whether something is so or not. All other aspects and characteristics of science can be understood directly when we understand that observation is the ultimate and final judge of the truth of an idea."

– Richard P. Feynman, The Uncertainty of Science (1963)

Scientific Temper

"It is the scientific approach, the adventurous and yet critical temper of science, the search for truth and new knowledge, the refusal to accept anything without testing and trial, the capacity to change conclusions in the face of new evidence, the reliance on observed fact and not on preconceived theory, the hard discipline of the mind – all this is necessary, not merely for the application of science but for life itself and the solution of its many problems."

"The scientific approach and temper are, or should be, a way of life, a process of thinking, a method of acting and associating with our fellow-men."

– Jawaharlal Nehru, The Discovery of India (1946)

Novel formulation: "scientific temper" as a "way of life".

Science versus Dogma

Science: evidence based, tentative/falsifiable, gather new knowledge.

Dogma (appearing as tradition or prejudice): unquestioning acceptance, absolute/unchanging, imposed by authority.

Conflict: When conclusions drawn by the scientific method do not agree with dogma.

- "*Eppur si muove* (and yet it moves)" – attributed to Galileo some time after he was forced to recant the heliocentric theory.

Caveat:

> *"Dogmatism and skepticism are both, in a sense, absolute philosophies; one is certain of knowing, the other of not knowing. What philosophy should dissipate is certainty, whether of knowledge or ignorance. "*
>
> – Bertrand Russell, The Problems of Philosophy (1912)

Rationality, Science, and Society

Rational thinking: Evidence based, questioning, critical thinking, non-acceptance of authority, opposed to dogma of tradition, occurs in various flavours.

- Scientific method is a specific type of rational thinking.
- Removing social prejudice requires rational thinking.

Caveat: Rationalism in a strict philosophical sense has a restricted meaning.
(See `https://plato.stanford.edu/entries/rationalism-empiricism/`.)

Outline of the Narrative: Rationalism/Modern Science, Dogma, and Reform in 19th Century Bengal

Rationalism and Social Reforms in 19th Century Bengal

Early to mid 19th century: Social and educational reforms.

- Social reforms - women's issues, social taboos/prejudices, ...
- Educational reforms - start of science curriculum, vernacular education, availability of basic science literature, new institutes.
- Social reformers called for *both* social reforms *and* science education with rationalism as the driving force.
  - Social reforms exhibit 'correlation' to growth of rationalism.
- Also, conservative defense of tradition, advocacy of *controlled* modernisation of education.

Late 19th century: Science-religion relation - unique viewpoints.

Novelty: Connects ideas across time, analytical insights, questions.
Caveat. Some connections are intuitive; need (in)validation.

19th Century Social Thinkers Included in the Narrative

Raja Rammohan Roy (1772–1833)

Raja Sir Radhakanta Deb (1784–1867)

Henry Louis Vivian Derozio (1809–1831) and the Derozians

Ishwar Chandra Vidyasagar (1820–1891)

Akshay Kumar Datta (1820–1886)

Bankim Chandra Chattopadhyay (1838–1894)

Swami Vivekananda (1863–1902)

The narrative does not discuss actual science and scientists.

Incompleteness of the Narrative

The narrative does not include all 19th century social thinkers: Reasons for non-inclusion are one or more of the following.

- Major work done in 20th century.
- Would not fundamentally change the narrative.
- Could not accommodate due to time constraints.

Intellectual health of 19th century Bengal: vibrant, teeming with new ideas. Now?

Raja Rammohan Roy: Rationalist Foundation and Institution Building

Rammohan Roy (1772–1833)

- Scholar: Bengali, Hindi, Sanskrit, Persian, Arabic, and English, later Hebrew and Greek; translated several Upanishads into Bengali, Hindi and English.
- Argued for the equal inheritance of property by women (1822).
- Anti-sati movement, leading to Bengal Sati Regulation (Dec 4, 1829): **First** strike of reason against dogma of tradition.
- Education: Founded Anglo-Hindu School (1822), advocated women's education, supported setting up of Hindu College (1817).
- Founded the Brahmo Samaj (1828) as a socio-religious reform movement.
  - Monotheistic, against idol worship, ...
  - Are the principles of Brahmo Samaj relevant today?

Rammohan Roy: Journalism

Founded/edited/associated: *Sambad Kaumudi* (1821), *Mirat-ul-Akhbar* (1822), *Brahmanical Magazine* (1821), *Bangaduta* (1829).

Licensing Regulations (1823) (gagging act) of the East India Company: need a license to publish.

- In response to criticisms by the Calcutta Journal.
- To assert control over newspapers in Indian languages.
- Pattern across time in how to gag critical voices?

Protest:

- Rammohan Roy closed *Mirat-ul-Akhbar* as a protest.
- Jointly filed a case in the SC – rejected; appealed to the King-in-Council – dismissed.

Press Freedom: Charles Metcalfe (1835)

Rammohan Roy: Views on Modern Education

Articulated in an 1823 letter to Lord Amherst.

- Note: Amherst Street in Kolkata was later renamed as Raja Rammohan Roy Sarani.

Context (quotations from the letter):

> *"... the government of England had ordered a considerable sum of money to be annually devoted to the instructions of its Indian subjects."*

> *"... the government is establishing a Sanskrit school under Hindu Pundits, to impart such knowledge as is already current in India."*

Rammohan Roy's Letter to Lord Amherst: Objections to the Proposal

Explosive (but needs to be seen in a broader context)

> *"... the Sanskrit system of education would be best calculated to keep this country in darkness"*

Difficulty:

> *"The Sanskrit language, so difficult that almost a lifetime is necessary for its acquisition, is well known to have been for ages a lamentable check on the diffusion of knowledge; and the learning concealed under this almost impervious veil is far from sufficient to reward the labour of acquiring it."*

Rammohan Roy's Letter to Lord Amherst: Objections to the Proposal

"... load the minds of youth with grammatical niceties and metaphysical distinctions of little or no practical use to the possessors or to society."

"... no improvement can be expected from inducing young men to consume a dozen of years of the most valuable period of their lives in acquiring the niceties of Baikaran or Sanskrit grammar."

Rammohan Roy's Letter to Lord Amherst: Objections to the Proposal

"Neither can much improvement arise from such speculations as the following, which are the themes suggested by the Vedanta, – in what manner is the soul absorbed into the Deity? What relation does it bear to the Divine Essence? ..."

"... no essential benefit can be derived by the student of the Mimansa from knowing what it is that makes the killer of a goat sinless on pronouncing certain passages of the Vedanta, and what is the real nature and operative influence of passages of the Vedas"

"The student of Naya Shastra cannot be said to have improved his mind after he has learned from it into how many ideal classes the objects in the universe are divided, ..."

Rammohan Roy's Letter to Lord Amherst: Suggestions

Plea to

> *"promote a more liberal and enlightened system of instruction; embracing Mathematics, Natural Philosophy, Chemistry, Anatomy, with other useful sciences"*

to be taught by

> *"a few gentlemen of talents and learning"*

and

> *"providing a college furnished with necessary books, instruments, and other apparatus."*

Rammohan Roy's Letter to Lord Amherst: Stress on Bacon

Compared the existing system of knowledge in India with pre-Baconian scholasticism (schoolmen) in Europe.

- Francis Bacon (1561–1626): Empiricism – early philosophy of science.
  - Science: *Inductive inference from observations (obtained from experiments) avoiding biases (called idols).*
  - Emphasis on observation/experimental evidence.
- Scholasticism – prevalent in pre-17th century Europe.
  - Unquestionable authorities: the Bible, Church fathers, Aristotle.
  - Inference based on deductive logic and linguistic analysis; spider's web of reasoning (as critiqued by Bacon).

Rammohan Roy's Letter to Lord Amherst

The petition was unsuccessful.

- **Importance:** Early call for govt. funding of science education.

Sanskrit College (1824); curriculum: Sanskrit and English.

- Early notable science alumnus: Madhusudan Gupta (1800-1856); transitioned from Ayurved to modern medicine, performed human dissection (breaking taboos) in 1836 at CMC.

Hindu College (1817); later *Presidency College* (1855) and government take over, admission to all communities.

- Early adoption of science curriculum.
- Brilliant alumni scientists of Hindu/Presidency College (mid/late 19th to early/mid 20th century):
  - R. N. Sikdar, J. C. Bose, P. C. Roy, M. N. Saha, S. N. Bose, P. C. Mahalanobis, and others.

Rammohan Roy's Letter to Lord Amherst: Broader Context

- Rammohan Roy was a Sanskrit scholar; he knew what content he did not wish students to spend time on.
- The petition was to initiate study of subjects which are useful for society.
- Some of the criticism of content were directed more at the philosophical questions rather than at the specific systems of thoughts.
- Argument against the use of government funds to set up a *new* Sanskrit 'seminary'; he suggested that promotion of Sanskrit learning can be done by "holding out premiums and granting certain allowances" to existing teachers.

Raja Sir Radhakanta Deb: Resistance to Rationalism/Social Reforms, Controlled Modernism

Raja Sir Radhakanta Deb (1784–1867)

Influential and wealthy: Part of the Sobhabazar Raj family.

Scholar: Along with Bengali and English, proficient in Sanskrit, Persian, and Arabic; published *Shabda Kalpadruma*, a Sanskrit language dictionary.

Promotion of education:

- Supported English education.
- Involved with Calcutta School Book Society (1817), Calcutta School Society (1818).
- Co-founded Hindu School.
- Wrote a book *Strisikhar Bidya* (1822) advocating women's education.

Raja Sir Radhakanta Deb (1784–1867)

Social views: Limited and *controlled* exposure to modern Western education.

Strongly conservative:

- Opposed Rammohan's effort to legislate against sati.
- Opposed Vidyasagar's effort to legalise widow remarriage.
- Active role in the dismissal of Derozio from Hindu College.

Raja Sir Radhakanta Deb (1784–1867)

Dharma Sabha: Founded by Radhakanta Deb in 1830.

- Immediate reason: to protest against the legislation to ban sati.
- Broader cultural context: “a society in defense of the Hindu way of life or culture” – David Kopf.
  - Revive public memory? (The best way to get a bad law repealed is to enforce it strictly – Abraham Lincoln; made in a different context.)
- According to David Kopf:
  - Earliest protonationalist movement in modern India.
  - Most were not Brahmins.
  - Opposed to “intrusive forms of Westernization”.
  - Western structure: executive body with president, treasurer, ...

Derozio and the Derozians: Radical Rationalism

Henry Louis Vivian Derozio (1809–1831)

Poet: published two books of poems.

1826: Teacher at Hindu College (at about 17 years of age).

- Led the “Young Bengal” movement.
- Inspired students to embrace rational thinking.
  - To think for themselves (‘sacred duty’), to question, to not accept anything blindly.
  - Formed The Academic Association (1828) - India’s first formal debating society - to allow free exchange of ideas.
  - Study books outside the prescribed curriculum.
  - Meet and discuss with students outside the prescribed hours.
- Left a lasting intellectual legacy.

Dismissal of Derozio from Hindu College

April 1831: Deliberations by a 9-member Managing Committee.

- Competence: Majority in favour of Derozio.
- Dismissal based on public feeling in the Hindu community: voted 6 to 1 in favour of dismissal; Hare and Wilson abstained.
  - Derozio was asked to resign, which he did.
  - Derozio was not provided an opportunity to defend himself.
  - According to Derozio, dismissal "without even the mockery of a trial."

Summary: Asked to resign by 'top' college without providing a hearing *not* for incompetence but due to public mood. Rings a bell?

Derozio's Rationalism

Excerpts from an 1831 letter to H. H. Wilson post 'resignation'.

- *"How is any opinion to be strengthened but by completely comprehending the objections that are offered to it ...?"*
  - Common in the Sanskrit philosophical tradition to state rival views clearly and precisely (K. K. Chakrabarti, 2010).
- *"Doubt and ignorance besiege us too closely, ... far be it for me to say "this is" "and that is not"."*
  - Feynman (1963) demands for future generation a "philosophy of ignorance" where "doubt is not to be feared, ... it is to be welcomed"
  - I should wish to preach the "will to doubt." – Russell (1922)
- *"... the highest wisdom assures man of his ignorance."*
  - Socrates: 'I know that I know nothing' is the highest wisdom.

Prominent Derozians

- **Radhanath Sikdar** (1813–1870)
- **Peary Chand Mitra** (1814–1883)
- **Ramgopal Ghosh** (1815–1868)
- **Krishna Mohan Banerjee** (1813–1885)
- **Ramtanu Lahiri** (1813–1898)
- **Dakshinaranjan Mukherjee** (1814–1878)
- **Rasik Krishna Mallick** (1810–1858)
- **Sib Chandra Deb** (1811–1890)

Impact of Derozians

- Radical rationalism: questioned scriptures and traditions; broke social taboos.
- Critical thinking: embraced empiricism, rejected blind adherence to religion.
- Women's causes: education, supported widow remarriage, opposed child marriage.
- Public expression of ideas: debates, speeches, essays.
- Writers, educationists, journalists.
- Adopted rationalism as a 'way of life'.
  - Resonates with Nehru's view of scientific temper as a way of life.
- Lasting intellectual impact, limited institutional legacy.

Ishwar Chandra Vidyasagar: Rationalism, Substance, Institutions

Ishwar Chandra Vidyasagar (1820–1891)

Social reformer

- Prime mover behind the Hindu Widows' Remarriage Act (Jul 26, 1856): **Second** strike of reason against dogma of tradition.
- Opposed polygamy, child marriage.

Educationist

- Curriculum revamp of Sanskrit College after he became Principal in 1851; dropped the 'Brahmin' admission criterion.
- Set up about 20 model schools and 35 girl's schools during his tenure as an Inspector of Schools.
- Championed education in vernacular language.
- Advocated "train the teachers" model.

Author: simple Bengali prose.

- Text books on Bengali and Sanskrit grammar; literary adaptations; books on social reforms (*Bidhababibaha*, *Bohubibaha*).

Ishwar Chandra Vidyasagar: Science Education

Jeevan Charit (1849):

- Short biographies of great men ("*mahatma*") including Copernicus, Galileo, Newton, and other scientists.
  - Tells the story of Galileo's struggle against dogma of tradition in his expounding of the heliocentric theory.
- The selection highlights Vidyasagar's emphasis on modern science.

Bodhoday (1851):

- One of the early textbooks in a vernacular language on the 'world around us' theme.
- Included topics on basic life science, geography, and also political economy ("*porishrom-adhikar*").

Vidyasagar-Ballantyne Debate

Sanskrit College Curriculum Review:

- Vidyasagar's proposal (1850-51).
- Ballantyne's report (1853).
- Vidyasagar's rebuttal of Ballantyne's report (1853).
- Vidyasagar's letter to Mouat (1853).

The three documents authored by Vidyasagar provide a glimpse into Vidyasagar's views on education.

Vidyasagar: Views on Education

Emphasis on Bengali:

- Grammar: start with “rules dressed in the easiest Bengali” and then move to Sanskrit.
- “Study of Bengali books, treating on useful and entertaining subjects be introduced” to the Junior Department.

Mathematics: Study of arithmetic, algebra, and geometry in Bengali, followed by Astronomy also in Bengali.

- Replace the Sanskrit texts *Lilavati* and *Vijaganita*: “needlessly difficult”, “examples are too few”.
- Calls for the compilation of Bengali books from “the best English works on those subjects.”
  - Such books will be of “great use also in the Vernacular Schools.”

Vidyasagar: Views on Education

English: Compulsory, started after some proficiency in Sanskrit.

Proposed study of Mill and Bacon as part of English studies.

- John Stuart Mill (1806–1873): A philosopher of science.
  - Emphasised observation, formulated Laws of Causality – a formalisation of inductive inference.
  - Liberal outlook, controversial views on colonialism.
- Proposal to teach scientific method to *general students*.
  - Resonates with Nehru's view of scientific temper as a way of life.

Vidyasagar: Views on Education

Strongly opposed the teaching of Berkeley's philosophy on the grounds of its similarity to Vedanta and Sankhya.

- George Berkeley (1685–1753): Pushed empiricism into metaphysics.
  - All knowledge comes only from sensory experiences; so only mind and ideas exists.
  - *Denied the existence of matter*.
  - Things exist only if they are perceived; posits God as constant observer to ensure continuity of the physical world.
- Both Vedanta and Sankhya are concerned with metaphysics; matter as illusion will be found to be similar to aspects of Vedanta.

Down-to-earth: Vidyasagar preferred Mill's epistemic empiricism over Berkeley's metaphysical empiricism.

- Recall: Criticism by Rammohan Roy regarding the uselessness of teaching metaphysical content of some of the Indian philosophical systems.

Vidyasagar: Overcoming Prejudice Through Education

"... they [students] have begun to think in such a way as to promise that hereafter every qualified student will be found free from all the prejudices of his countrymen."

Vidyasagar: Towards Mass Education

"What we require is to extend the benefit of education to the mass of the people. Let us establish a number of vernacular schools, let us prepare a series of vernacular class-books on useful and instructive subjects, let us raise a band of men qualified to undertake the responsible duty of teachers and the object is accomplished."

Interlude: ‘Everything in the Shastras’ Seesaw

Everything in the Shastras: Reason Strikes Dogma

1853: Vidyasagar on "*The bigotry of the learned of India*":

> *"They believe that their shastras have all emanated from the omniscient Rishis and, therefore cannot but be infallible."*
>
> *"... when they hear of a scientific truth, the germs of which may be traced out in their shastras, instead of showing any regard for that truth, they triumph and the superstitious regard for their own shastras is redoubled."*

Everything in the Shastras: Reason Strikes Dogma

1939: Meghnad Saha (in *Adhunik Bijnan O Hindu Dhormo ('Sobei Baade Achche')*):

> *"bigoto kuri botsore bed, upanishad, puran, ittadi somostyo hindushastrogronthyo ebong hindu jotish o aprapor bigyan sombhondhiyo gronthadi tanyo tanyo koriya khujiya ami kothao abishkar korite sokhkhom hoi nai je, ei somostyo prachin gronthye bortomona bigyaner multotyo nihito achche"*
>
> *"dukhyer bishoy deshe eirup apobigyan procharoker abhab nai, tahara sotyer naame nirjola mithyar prochar koritechen matro"*

Everything in the Shastras: Dogma Strikes Back

Public claims by public figures:

- **2014:** Genetic science existed in ancient India.
- **2014:** Nuclear tests were conducted lakhs of years ago in India.
- **2018-2019:** Theory of evolution is incorrect, since it is not in ancient Indian texts.
- ...

https://www.thehindu.com/features/metroplus/saturday-spin-when-everything-came-from-india/article6552177.ece

https://caravanmagazine.in/science/false-scientific-claims-modi-first-term

Pushback: By various science organisations, newspapers articles, ...

Do we keep seesawing forever?

Akshay Kumar Dutta: Rationalism, Science Writing, Society through Rationalism

Akshay Kumar Dutta (1820–1886)

Pioneering Bengali science writer and scientific vocabulary.

- Wrote early (first?) vernacular textbooks: *Bhugol* (1841) and *Padarthavidya* (1856).
- 3-part textbook *Charupath* (1853, 1854, 1859) covering geography, physics, earth sciences, life sciences, technology (printing press, telegraph, steam engine), rationalism/empiricism ("prakitir mandir"), morality.

Akshay Kumar Dutta (1820–1886)

Editor of *Tattwabodhini Patrika* (established by Debendranath Thakur in 1843).

- Wrote articles on geography, physics, and astronomy;
- Column *Palligramastha Prajader Durabastha Barnan* on socio-economic suffering; criticised permanent settlement.
- Women causes: widow remarriage, abolish polygamy, ...

Books:

- *Bahya Bastur Sahit Manavprakritir Sambandha Vichar* (1851, 1853).
- *Dharmaneeti* (1856): on the principles of morality.
- *Bharatvarshiya Upasak Sampraday:* A 2-volume work on religious anthropology.

Akshay Kumar Dutta (1820–1886)

Outlook:

- Detailed investigation of rational explanations for human behavior, ethics, and health.
  - This sets him apart from prominent 19th century rationalists.
- Emphasised rational/scientific thoughts over superstitious beliefs.
- Advocated empiricism; challenged the infallibility of the Shastras.

Reportedly used the following 'math' in verbal arguments:

labour = crops; labour + prayer = crops; so prayer = 0.

Bankim Chandra Chattopadhyay: Support for Institutional Science, ‘Scientific’ Analysis of Religion

Bankim Chandra Chattopadhyay (1838–1894)

Sahitya Samrat:

- A very successful and influential novelist.
- Journalist: founded *Bangadarshan* and published many articles.
- Essayist: wrote on a variety of topics, including science.
  - *Samya:* investigates various forms of social inequality.
- Author of *Vande Mataram*, the national song of India.

Bankim Chandra Chattopadhyay: Women's Rights

Expert view: Amiya P. Sen writes (referring to *Dharmatattwa*):

> *"... in the late 19th century Bengal he was one of the staunchest opponents of both widow remarriages and the legal abolition of multiple marriages among men."*

`https://www.sahapedia.org/bankimchandra-chattopadhyay-and-renascent-bengal`

Counterpoint: The message of the essay *Samya* does not appear to support such a strong statement.

Bankim Chandra Chattopadhyay: Women's Rights

***Samya*:** Section 5 of the essay is a brilliant defense of women's rights.

- Penetrative rationalist style questioning of ethics, tradition, beliefs, and Shastras in support of women's rights.

Formulates four major issues:

- **Women's education:** whole hearted support.
- **Restrictions on women's freedom of movement:** unqualified condemnation.
- **Widow remarriage:** unstinted support for legal right.
  - Puts moral block: A wife who truly loved her previous husband will never remarry even if it is legally permitted.
- **Polygamy:** No moral basis for legal right to polygamy; also strongly condemns polygamy in another essay *Bahubibaha*.
  - Points to possible exceptions such as no heir, sick wife; but points out that if such exceptions are permitted, then they should also apply in reverse.

Bankim Chandra Chattopadhyay: Scientific Outlook

Support to institutional science: One of the early supporters of Mahendra Lal Sircar's ground breaking effort to establish the Indian Association for the Cultivation of Science.

Spreading established scientific knowledge: Early pioneer of popular science writing in a vernacular language.

- Essays on contemporary science written in Bengali.
  - Initially published individually in *Bangadarshan*; later published as *Bigyan Rahasya* in 1875.
- Included essays on solar eruption, multitude of stars, aerostation, universe, antiquity of man, protoplasm, moon.

Bankim Chandra: Analysing Religion Using Science

***Trideb Sombhondhye Bigyanshastra ki bole*.**
Primary conclusion: The Trinity concept of creator, preservor and destroyer in Hinduism is neither unjustified nor against proofs.

> *"atoab sroshta, pata, ebong horta prithok prithok chaitanya, amot bibechona kora asongoto ebong promanbirudhyo nohe"*

Further remarks in the essay.

- Analytical validity of public belief in a firm innate basis for Trinity.
- Science does not prove the existence of Trinity; puts religious beliefs outside the scope of science.
- Trinity cannot be considered to be physical beings ("*sakar*"). "*Brahma, Bishnu, Maheswar prottekei kotokgulo adbhut upyanasher nayak. sei sokol upyanasher tilmatro noisorgik bhitti nai*"

Bankim Chandra: Analysing Religion Using Science

Trideb Sombhondhye Bigyanshastra ki bole

Methodology of the essay:

- Analysis based on Mill with occasional references to Darwin.
- Crucial for any scientific investigation: *verifiable observation*.
  - Not clear in the essay.
- An example of what Feynman called ‘cargo-cult’ science?

Swami Vivekananda: Unique Weaving of Science and Religion

Swami Vivekananda (1863–1902)

Presented Hinduism to the West:

- *Respectability:* A modern, philosophically rich "world religion".
- *Universalism:* Dictum by Shri Ramakrishna, *jato mat tato path*.
- *Counterbalance:* Spirituality versus technological advances.

Founded the Ramakrishna Math/Mission:

> *"This Math that we are building will harmonise all creeds, all standpoints. Just as Shri Ramakrishna held highly liberal views, this Math too, will be a center for propagating similar ideas. The blazing light of universal harmony that will emanate from here will flood the whole world."*

– Attributed to Vivekananda (`https://belurmath.org/about-us/`)

Is the world flooded with the blazing light of universal harmony emanating from the RKM?

Vivekananda: Helping Science

Letter from Jamsetji Tata to Vivekananda in 1898.

- Recollects Vivekananda's "views on the growth of the ascetic spirit in India, and the duty, not of destroying, but of diverting it into useful channels."
  - Novel position: Diversion of ascetic spirit into useful channels.
- States that Vivekananda would be the most suitable person to assume leadership of a Research Institute of Science for India which he wanted to establish.
  - Vivekananda did not take up the role.

Meeting with J. C. Bose in Paris in 1899-1900.

- Vivekananda convinced the reluctant J. C. Bose to file for a patent.
- The patent ("Detector for Electrical Disturbances") was filed in the US in 1901 through Sara Bull, who was a disciple of Vivekananda.

Vivekananda on Unity, the Goal of Religion

"Chemists are seeking to ... find one element that is the source of all other elements. Reaching that, they can go no further; the science of chemistry will have become perfect. So it is with the science of religion. If we can discover this perfect unity, there cannot be any further progress."

"Hindus ... regard religion and philosophy as but two aspects of one thing which must equally be grounded in reason and scientific truth."

"... search for final unity which is alike the goal of science and religion."

- Points out similarities between the goals of science and religion.
- Unity is the goal of science - needs further examination?.

Vivekananda on Science and Religion

"Experience is the only source of knowledge."

"... men who teach religion from experience ... are called mystics"

"... mystics in every religion speak the same tongue and teach the same truth. This is the real science of religion."

"... science of religion is based on the analysis of the human soul. It has no creed."

Religion "deals with the truths of the metaphysical world", "the book within".

Natural sciences "deal with the truths of the physical world", "the book without".

- Science of religion: replaces verifable experiments by experience.
- Draws boundary between religion and natural sciences - do natural sciences have a boundary?

Narrative Summary

Rationalism versus Tradition

Rammohan, Derozians, Vidyasagar: Early to mid 19th century.

- Advocated social adoption of rational thinking and (radical) rejection of dogma of tradition.

Radhakanta Deb: Early to mid 19th century.

- Strong defender of tradition; opposed rationalism; advocated *controlled* modern intellectual influence.

Ideology/Idealism versus Empiricism/Ground Reality:

- Sati: Divine and voluntary (Radhakanta Deb) vs. graphic description of the way it was actually done (Rammohan Roy).
- Widow remarriage: Duty and scriptural adherence (Radhakanta Deb) vs. real pitiable condition (Vidyasagar).

Rationalism versus Tradition

Akshay Kumar Dutta. mid 19th century.

- Bridge figure: Along with advocating rationalism, initiated the study of social/ethical problems from a rationalist viewpoint.

Bankim Chandra and Vivekananda: Late 19th century.

- Bankim Chandra: Supported institutional science; spread estabished scientific knowledge; attempted scientific analysis of religion.
- Vivekananda tried to identify a common goal of science and religion; attempted to explain religion as science.

Taboos/Tradition versus Shastra: Another Dimension

Primary arguments for social reforms: rational;
Secondary supporting arguments from Shastras.

- Rammohan: women's inheritance; opposition to sati.
- Vidyasagar: widow remarriage.
- Corpse dissection in 1835-36.

Interdependence of arguments (simplified view):

- Rammohan argued *Manusmriti* dictated that widows live an ascetic life, hence no sati; but, also no remarriage;
- Vidyasagar (later) argued that *Manusmriti* applied to *Satya Yuga*; in *Kali Yuga*, the *Parashara Samhita* applies which permits widow remarriage.
  - Applicability of *Manusmriti* vis-a-vis *Parashara Samhita* possibly relevant to other contexts too.

Requirement for secondary arguments: To convince tradition soaked society; British had a hands off policy, no action unless call for reforms rose from within the society.

Tradition vs. Rationality Debate of Current Relevance

Sabarimala Temple and related issues.

- Tackling the rationalist argument: rebut or go around?
- How do the recent arguments on this issue compare with arguments forwarded by conservatives in 19th century Bengal?

Growth of Modern Science: Catching Up

19th century realisation about modern science (and technology): Vehicle of world domination by the West; it is essential to catch up.

- As social thinkers, science was viewed as useful (contrasts with the view that scientists study nature not because it is useful, but because it is beautiful – Henri Poincaré).

Growth of science: Factors.

- Early to mid 19th century: science curriculum, availability of vernacular science (text)books, growth of printing press, questioning social taboos, rationalist outlook.
- Mid to late 19th century: institutionalised science, government funding of science education.
- Colonial education policy: long history throughout 19th century.
- Economic factors.

Growth of Modern Science: Catching Up

Science up to mid 20th Century: sparse, but brilliant!

Later: Are we still playing the catch up game?

- Growth of institutional science.
- Is a uniform general scientific temper necessary in the society to achieve great science? How about the US?
- China and Japan (as Asian examples) started the catch up game also in the 19th century. How do we compare?
  - Economic conditions are different.

Motivation and Conclusion

Is the Present Discussion a Waste of Time?

Motivations:

- **Article 51A(h) of the Constitution:** It shall be the duty of every citizen of India to develop the scientific temper, humanism and the spirit of inquiry and reform.
- **The Hollywood film Coco as a parable:** An idea 'lives' if someone in the living world remembers it.
- **Cultural appropriation:**
  - "The devil can cite Scripture for his purpose." – William Shakespeare.
  - "*tomar pujar chole tomaay bhulei thaaki*" – Rabindranath Thakur.
- **Hans Christian Andersen:** Call out the Emperor's New Science, and appreciate **actual** ancient Indian science.
- **Hirak Rajar Deshe:** "*era joto beshi pore toto beshi jaane, toto kom maane*." – Satyajit Ray.

Last Comments

Reasoning versus Believing: Believing is easier, more (intellectually) energy efficient; energy efficiency is an evolutionary adaptation.

- If everybody believes the same things, then society is uniform.
- If individuals reason for themselves, they may come to different conclusions – same procedure, but different contexts.

Can reasoners 'hang together'? (We must, indeed, all hang together or, most assuredly, we shall all hang separately – attributed to Benjamin Franklin (1776); made in a different context.)

Pedestalising: Humans become super-humans/divine; their words become gospel/authority.

- The criticism can be applied to the quotations in this presentation.

Circularity: If you ask people to question, then those people may in turn question you.

Prarthana

Chitto jetha bhaysunyo, uchcho jetha shir
...
jetha tuchcho acharer morubalirashi
bicharer stotropoth phele nai grashi
...
nijo hoste nirdoy aghat kori prito
bharatere sei sorgye koro jagorito

– Rabindranath Thakur (1910)

Reason/Tradition tension: "Where the clear stream of reason has not lost its way into the dreary desert sand of dead habit"

Prayer: "Where the mind is without fear, and the head is held high; ... Into that heaven of freedom, my Father, let my country awake."

Is the poem suitable as a prayer in schools?

Thank you for your kind attention!

“laharir por lahari tuliya
aghater por aghat kor”

– Rabindranath Thakur, *Nirjhorer Sapnobhangyo* (1882)

References

Digital versions of the primary references are scattered across the net.

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